



REPORT WRITING

Technical and Business Writing

Assignment 2



GROUP MEMBERS

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TBW SPRING-25
DEPARTMENT OF COMPUTER SCIENCE
ASSIGNMENT-2

TOPIC: REPORT WRITING

Total Marks: 50

The assignment will be done in pairs. You will stick to the pairs made for assignment 1. Only 1 member of the pair will submit the assignment.

Instructions:

You will be presented with three (3) scenarios related to short informal reports. For the first two scenarios, answer the questions provided in **100-150 words** each. You will write a complete report in response to the third scenario.

Ensure that your responses are clear, concise, and adhere to the following guidelines:

1. **Reader-Friendly Structure:** Organize your answers in a way that makes it easy for the reader to follow. Use headings or bullet points where necessary to improve readability.
2. **Appropriate Tone:** Make sure your answers are written in a professional tone, keeping in mind the specific audience of each report.
3. **Clarity and Conciseness:** Avoid unnecessary jargon and lengthy explanations. Stick to the essential details, presenting them in a straightforward manner.

Scenario 1: Incident Report (10 Marks)

Scenario:

You are a software developer at XYZ Tech Solutions. During the final testing phase of a new e-commerce website, a critical bug was discovered that causes the shopping cart to malfunction. Customers are unable to add or remove items, leading to potential loss of sales. The bug appears after the latest code deployment, which integrated a third-party payment gateway.

Questions:

1. What are the key components you would include in the incident report to ensure all necessary details about the bug are captured? (5 Marks)

Ans: The incident report of the shopping cart malfunction should include:

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- Short summary including description of the issue about the problem that occurred and what were its effects and where did it occur.
 - The exact date and time when the issue was observed.
 - Brief description of the bug that occurred.
 - What part of the code was deployed which caused the issue to arise.
 - What parts of the website were affected by the bug (add and remove functionalities of shopping cart)
 - Steps to reproduce the bug in an isolated environment to understand the problem better.
 - Who was involved in the incident.
 - Specific teams to be informed.
 - Graphics and data analysis of the financial impacts, customer complaints and failed transactions.
 - Actions taken immediately to handle the situation.
 - The status of the issue whether the issue has been resolved, under investigation or in the process of resolving.
 - Recommendations for the future to avoid such incidents in the future.
2. How would you prioritize this incident, and what immediate steps would you recommend to resolve the issue? Consider aspects like team communication and testing environments. (5 Marks)

Ans: I would recommend the following steps to resolve the issue:

- Roll back the latest deployment to restore the functionality and minimize the financial loss.
- Contact and coordinate with the development, quality assurance and customer support teams to ensure effective resolution.
- Reproduce the bug in an isolated environment to analyze the part of code which was root cause and understand the problem better without affecting current users.
- Debug and fix the faulty code.
- Conduct testing of the code and ensure its compatibility with other functionalities of the website before redeploying the code into the actual code.
- Monitor the user activity and error reports to verify that the issue is resolved successfully.
- Conduct a post-mortem review to avoid such incidents in the future.

Scenario 2: Trip Report (10 Marks)

Scenario:

You attended the "2025 International Conference on AI and Machine Learning" held in Silicon Valley. The conference lasted for 3 days and featured keynotes, technical sessions, and hands-on workshops. Your manager has asked you to prepare a trip report summarizing

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the key takeaways, and any recommendations on how the knowledge gained could benefit the company's AI development projects.

Questions:

1. What sections would you include in your trip report, and what information should each section contain? (5 Marks)

Ans: The following sections should be included in the trip report:

- **Introduction:**
 - **Purpose:** Explain the reasons for attending the conference.
 - **Objectives:** Mention specific goals such as learning new AI technologies and networking with emerging industry.
 - **Personnel:** List the company representatives who attended the conference with you along with their positions.
 - **Authorization:** Mention who approved your application for the trip and approved of the budget and resource allocation for the trip.
 - **Discussion:**
 - **Conference Organizers and Hosts:** List of the hosts, organizers and sponsors of the conference.
 - **Key Points of Sessions:** A summary of valuable notes, technical talks and the workshops attended.
 - **New Technologies Introduced:** Outline the emerging AI technologies introduced.
 - **Contacts Made:** Enlist the people representatives of the companies you met and connected with.
 - **Relevance to Company's Projects:** Explain how the conference will help in enhancing the company's ongoing projects.
 - **Challenges Encountered:** Mention any kind of obstacles faced including time constraints, lack of resources, etc.
2. How would you highlight the most relevant insights gained from the conference, and what actionable recommendations would you make for the company's AI initiatives? (5 Marks)

Ans:

Highlighting Insights from Conference:

For highlighting the most relevant insights gained from the conference, focus on:

- **Emerging AI advancements** such as Improved deep learning, advanced AI model training, etc.
- **Technologies and tools** that can be integrated into current projects.
- **Best techniques** shared by the conference leaders to improve efficiency.
- **Market trends** that could benefit the company in AI development.

Recommendations:

- **Adopt New AI Techniques:** Implement emerging techniques to enhance model and accuracy.
 - **Explore AI Automation Tools:** Keep looking for new AI tools to compete with the rising industries.
 - **Enhance AI Ethics:** Improve fairness and data security in AI models.
 - **Education of Employees:** Conduct internal training sessions to make sure employees are aware of sufficient knowledge to use emerging technologies.
 - **Collaboration and Networking:** Expand company connections to explore partnerships and collaboration opportunities.
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Scenario 3: Feasibility Report (30 Marks)

Scenario:

Your company, Digital Innovators Inc., is considering developing a mobile application designed to help small businesses manage inventory and sales data more efficiently. As part of the planning phase, you have been tasked with preparing a feasibility report to evaluate whether this project presents a viable business opportunity. In your report, you should examine factors such as market demand, technical requirements, development costs, and the competitive landscape.

Task:

Write a feasibility report that assesses the viability of developing the mobile application. Your report should provide a detailed analysis of the following aspects:

1. **Main Sections of the Feasibility Report:** Identify and describe the key sections that should be included in the report. For each section, explain the purpose and the type of information it should contain.
 2. **Risk Assessment:** Evaluate the potential risks associated with developing the app. Focus on the challenges related to market competition and technical limitations, and propose strategies to mitigate these risks.
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FROM: Wukong, Marketing Manager
TO: Eeman Mirza, CEO, Digital Innovators Inc.
DATE: March 12, 2025
SUBJECT: Feasibility for inventory management application

Purpose of the report

The purpose of this report is to study whether the project, a mobile application catered towards managing inventory and sales data, is a viable business opportunity in the contemporary market landscape. After analyzing its technical feasibility, market scope, development costs and its competitors, we will determine if this project presents a strong business opportunity.

Market analysis:

The global inventory management software market was estimated to be \$5.3 billion in 2023 and is estimated to grow at a steady rate of 8% annually for the next six years. Small businesses account for 44% of this market share, making them a significant segment. While the market does look attractive, the competition is fierce. There are a multitude of companies competing for market share.

On the other hand, Pakistan's inventory management software market, while lesser in comparison, has a lot of potential to grow. Most small businesses, like utility stores, still manually bookkeep their sales and inventory data. Given the current global technological revolution trend, this market is bound to grow.

Currently, there are two main companies competing in Pakistan for this market:

1. **QuickBooks** – An internationally renowned bookkeeping software known for being feature-rich and versatile. It is complex and lacks support for the Urdu language.
2. **DigiKhata** – A start-up that has laid strong foundations in this market and accumulated a user base of over 5 million users. While it is user-friendly in comparison to QuickBooks, it lacks modern AI-powered solutions. In addition, while support for the Urdu language is provided, it does not support regional languages such as Pashto, Punjabi, or Sindhi.

Given these gaps in the market, Digital Innovators Inc. has a strong opportunity to introduce a localized, AI-powered, and easy-to-use inventory management solution tailored for Pakistani SMEs.

Technical Feasibility:

This mobile application requires key technical decisions:

Platform Selection:

The app should be launched on both iOS and Android, as these are used by the majority of Pakistani small business owners.

Framework for Development:

There are three frameworks we can consider:

- **Lynx** – A new framework introduced in 2025, known for its high development and runtime speeds, which can reduce long-term costs.
- **React Native** – An older but stable development option. However, it requires separate development for iOS and Android, effectively doubling costs.
- **Flutter** – Allows a single codebase for both iOS and Android, reducing development time and costs.

While Lynx is suitable in the long run, it is a new framework. Therefore, it is subject to a lot of bugs and our software engineers are not yet proficient in using this framework. Furthermore, since we want to create an MVP first and ship our product as soon as possible. Flutter is best suited for now. Once the application's user base grows, we can consider migrating to Lynx.

Storage:

- **Self-hosted database** – Lower operational costs but requires significant initial investment. The maintenance time and costs are very high as well.
- **Cloud storage** – Higher operational costs and an initial investment is not required. Also, the maintenance efforts are significantly reduced, which improves reliability.

Due to its reliability, **cloud storage** will be chosen.

Security:

A meeting with our company's cybersecurity team has already been conducted, and they have confirmed their capability to handle the application's security requirements.

AI-Powered Solutions:

- **Training our own models** – Expensive and requires expertise our company currently lacks.
- **Fine-tuning existing models** – We can integrate OpenAI models alongside Retrieval-Augmented Generation (RAG) to enhance performance.

Here is a summary of the decisions:

Aspect	Decision	Justification
Platform	iOS & Android	Majority of Pakistani small business owners use these OS.
Development Framework	Flutter	Single codebase for both platforms, reducing time and cost.

Storage	Cloud Storage	More reliable
Security	Handled by in-house team	The cybersecurity team has confirmed capability.
AI Solutions	Fine-tune existing models (OpenAI + RAG)	Cheaper and feasible compared to training our own models.

Financial Analysis:

1. Cost Analysis

Developing and maintaining an application requires several costs. Here is an estimate:

Cost Component	Estimated Cost (PKR)
App Development (MVP)	14,000,000
Backend Infrastructure	4,200,000
AI-Powered Features	5,600,000
UI/UX Design	2,800,000
Testing & Quality Assurance	1,400,000
Total Development Cost	28,000,000
Annual Maintenance Cost	5,600,000

2. Revenue Model:

The most widely used model in this market is the freemium model. This model is perfect for Pakistan, as most Pakistanis are unwilling to pay initially. Once mutual trust is achieved, we can offer attractive features that are only available through paid subscriptions.

Given the economic condition of Pakistan, the pricing should be kept as follows:

Basic Plan: Free

Standard Plan: PKR 2,800/month

Enterprise Plan: PKR 8,400/month

Since we lack funds to choose a cost leadership strategy (win market through very low prices), we will opt for a focused differentiation strategy (selling uniqueness – in our case AI solutions and regional language support) and utilize it to break even on our investments as soon as possible.

Competitive Landscape:

Risk Analysis:

- **Market Saturation** – Well-known apps like Quickbook and DigiKhata already dominate the market which makes customer acquisition challenging.
- **Pricing Pressure** – Companies with excess funds like Quickbook, are able to follow a cost leadership strategy, hence acquiring customers through low prices
- **Early-bird advantage** – Early applications like DigiKhata have multiple relationships with banks, market unions and other companies.

Mitigation Strategies:

- **Focused Sales Approach Focused Sales Approach** – Educating small business owners who use manual data collection methods about the benefits of a software solution will no longer be required as this task will have already been completed by large companies. Instead, our sales strategy will focus on selling our product directly.
- **Unique Selling Proposition** – Instead of relying on a pricing strategy, we will gain market share through our cutting-edge Ai powered solutions.
- **Befriend indirect competitors** – Instead of trying to form relationships with the same banks and companies, we are better off collaborating with the competitors of these banks and companies.

Technical Limitations:

Risk Analysis:

- **Data Security & Compliance** – The security of our user's data should be our main priority. While our team of cyber security experts can initially manage our applications, however, as our user base increases the security risks will increase.
- **Scalability Issues** – As the application grows in popularity, server load and slow response time may occur.
- **Currency Devaluation** – Our company will need to pay the server and API costs in dollars since most of these technologies are not locally available. This means that the prices can skyrocket in case of currency devaluation.

Mitigation Strategies:

- **Outsourcing** – Once revenue is generated, we should outsource our data security maintenance to an IT service company that specializes in this field.
- **Scalable Cloud Infrastructure** – Take advantage of special features of cloud service providers like AWS or Microsoft Azure.

- **Offshore company** – Many services exist that allow foreigners to register a company in another country. We should store funds in another country where the currency would work in our favor.

Summary of the report:

This report evaluates the feasibility of launching a mobile inventory management application mainly in Pakistan. It analyzes market potential and competitiveness, technical requirements and limitations, and financial viability. Competitors like DigiKhata and Quickbooks were analyzed.

Recommended Actions:

Excluding the mitigation strategies mentioned, there are a few more suggestions.

- Target, initially, only the Pakistani markets. Once sufficient control is gained, launch this application in other countries.
- Incorporating regional languages and cultural difference into the application will be fruitful as this is a gap in the market
- The marketing of this product should be focused on door-to-door salesman rather than generalized online advertisement.

Submission Guidelines:

- The assignment should be submitted in a Word or PDF document.
 - Ensure your responses are clear, well-organized, and free of errors.
 - Use professional language and follow proper report-writing conventions.
 - Deadline for submission: [Insert Deadline]
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Wukong

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Marking Rubric:

Criteria	Marks Allocation
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Total Marks	50 Marks
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Scenario 1: **10 Marks**

Scenario 2: **10 Marks**

Scenario 3: **30 Marks (Rubric Attached below)**

Consider:

Answer clarity and relevance

Appropriateness of the report format

Depth of analysis and problem-solving

Structure and coherence

Rubric for Assessing Scenario 3 (Feasibility Report):

Criteria	Excellent (5-4 Marks)	Good (3 Marks)	Needs Improvement (2-1 Marks)	Unsatisfactory (0 Marks)
Structure and Organization	The report is well-structured with clear, logical sections. All required sections are present and well-organized.	The report has a logical structure, but some sections may lack clarity or organization.	The report lacks proper structure; some sections are missing or poorly organized.	The report lacks structure and does not follow the expected format.
Content Completeness	All sections thoroughly cover the relevant aspects of the feasibility analysis, including a deep dive into market demand, costs, technical	Most sections are addressed in sufficient detail, but some areas may require more analysis.	Some key aspects of the feasibility analysis are underdeveloped or missing.	The report lacks significant content and does not address key points.

Criteria	Excellent (5-4 Marks)	Good (3 Marks)	Needs Improvement (2-1 Marks)	Unsatisfactory (0 Marks)
	requirements, and competition.			
Risk Assessment	A comprehensive evaluation of risks related to competition and technical limitations, with clear mitigation strategies.	Adequate identification of risks and mitigation strategies, but some key risks or strategies may be missing.	Limited discussion of risks; mitigation strategies are vague or not clearly explained.	No meaningful discussion of risks or mitigation strategies.
Clarity and Writing Quality	The report is clearly written, free from errors, and easy to understand. Proper grammar and formatting are used.	The writing is generally clear, with minimal errors that do not affect understanding.	The writing is unclear at times, with several errors that hinder understanding.	The report is poorly written, with numerous errors that make it difficult to understand.
Analytical Depth	The analysis is thorough, demonstrating strong critical thinking and a deep understanding of the business opportunity.	The analysis is solid, though may lack depth in some areas or critical thinking.	The analysis is shallow or incomplete, missing key insights or important points.	The analysis is superficial or completely lacking.

Note to Students:

Be sure to apply what you've learned in class about the structure and components of short informal reports. Make sure that your reports are clear, professional, and reflect a real-world approach to technical writing.